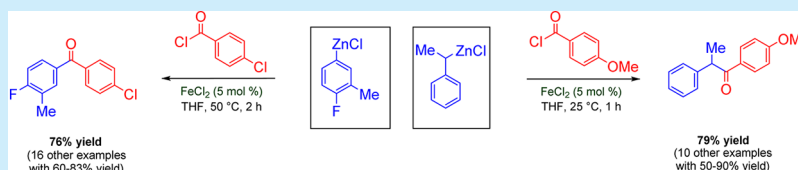


Iron-Catalyzed Acylation of Polyfunctionalized Aryl- and Benzylzinc Halides with Acid Chlorides

Andreas D. Benischke, Marcel Leroux, Irina Knoll, and Paul Knochel*

Department of Chemistry, Ludwig-Maximilians-Universität, Butenandtstrasse 5-13, 81377 Munich, Germany

S Supporting Information



ABSTRACT: FeCl_2 (5 mol %) catalyzes a smooth and convenient acylation of functionalized arylzinc halides at 50 °C (2–4 h) and benzylic zinc chlorides at 25 °C (0.5–4 h) with a variety of acid chlorides leading to polyfunctionalized diaryl and aryl heteroaryl ketones.

Organozinc reagents belong to the most useful organometallics for organic synthesis since they show a high tolerance of functional groups and undergo efficient transmetalations to various metallic salts leading to a broad reactivity pattern.¹ The reaction of zinc reagents with acid chlorides is an important method for preparing a range of polyfunctionalized ketones.² Although such acylations proceed without catalysts, the yields are usually low due to extensive side reactions.³ A transmetalation of organozinc reagents to the corresponding copper reagents using $\text{CuCN} \cdot 2\text{LiCl}$ ⁴ considerably extends the reaction scope of this acylation procedure and allows the preparation of various highly functionalized ketones.⁵ Although this acylation can be performed in many cases using substoichiometric amounts of copper catalyst, lower yields generally result under such conditions. Alternatively, Negishi⁶ reported a palladium-catalyzed acylation method of organozinc halides, and Rieke⁷ described a related nickel-catalyzed acylation for the preparation of aryl ketones. Recently, Gosmini⁸ reported a cobalt-catalyzed acylation of arylzinc bromides. Interestingly, organozinc reagents can also be acylated with thioesters as shown by Fukuyama⁹ or with acyl cyanides.¹⁰ Since iron salts are cheap, environmentally friendly, and of moderate toxicity, their use as acylation catalysts would be especially attractive.¹¹ Marchese¹² and Fürstner¹³ already demonstrated that iron-catalyzed reactions of acid chlorides with organomagnesium reagents can be readily performed. In addition, Cahiez¹⁴ showed that organomanganese reagents prepared from the corresponding Grignard species undergo acylation reactions with excellent yields.

Herein, we report an efficient iron-catalyzed acylation of polyfunctionalized aryl- and benzylzinc halides with acid chlorides. Thus, benzylzinc chloride (**1a**, 1.0 equiv), prepared by the insertion of Mg turnings and ZnCl_2 , reacts with 4-chlorobenzoyl chloride (**2a**, 0.8 equiv) in THF (25 °C, 0.5 h) in the presence of various transition-metal salts (Table 1).^{15a} This rapid screening indicated that iron(II) chloride is an excellent

Table 1. Screening of Catalysts for the Acylation of Benzylzinc Chloride (1a**) with 4-Chlorobenzoyl Chloride (**2a**)**

entry	catalyst	yield ^a (%)	entry	catalyst	yield ^a (%)
1		74	5	FeBr_3	85 ^b
2	MnCl_2	60	6	$\text{Fe}(\text{acac})_2$	73
3	CrCl_2	65	7	FeBr_2	82 ^b
4	$\text{Fe}(\text{acac})_3$	60	8	FeCl_2	90 ^c (70) ^d

^aIsolated yields of pure product. ^b FeBr_2 and FeBr_3 of 98% purity. ^c FeCl_2 of 99.5% purity. ^d FeCl_2 of 98% purity.²⁵

catalyst (Table 1, entry 8) in THF as solvent.¹⁶ Additives like 4-(dimethylamino)pyridine,¹⁷ $\text{Sc}(\text{OTf})_3$,¹⁸ InCl_3 ,¹⁹ $\text{BF}_3 \cdot \text{OEt}_2$,²⁰ and $\text{LaCl}_3 \cdot 2\text{LiCl}$ ²¹ did not lead to an improvement.²² In addition, the testing of various solvent mixtures showed that THF was the ideal solvent for this acylation reaction since the ketone **3a** was obtained after 0.5 h reaction time in 90% isolated yield.²³ To ensure that FeCl_2 is the true catalyst of this acylation, we have used FeCl_2 of high purity (99.5%).²⁴ The coupling experiment with FeCl_2 of lower purity (98%) proceeded somewhat less cleanly and gave 70% yield of product.

This iron-catalyzed acylation proved to be quite general. Thus, the reaction of (3-trifluoromethylbenzyl)zinc chloride (**1b**) with 4-*tert*-butylbenzoyl chloride (**2b**) provides within 0.5 h at 25 °C the desired ketone **3b** in 65% yield (Table 2, entry 1). In the absence of an iron catalyst, the yield leveled at 38% yield. Similarly, the acylation of **1b** with 4-chlorobenzoyl chloride (**2a**) gave the ketone **3c** in 71% yield (entry 2). Furthermore, (3-fluorobenzyl)zinc chloride (**1c**) reacted in the presence of FeCl_2

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Table 2. Iron-Catalyzed Acylations of Benzylic Zinc Reagents (1b–g) with Acid Chlorides (2a–e)

$\text{FG}_1 = 3\text{-CF}_3, 3\text{-F}, 3\text{-CO}_2\text{Et}, 4\text{-OMe}, 4\text{-SMe}$
 $\text{FG}_2 = 4\text{-F}, 4\text{-Cl}, 4\text{-}^t\text{Bu}, 4\text{-OMe}$

entry	benzylic zinc reagent	acid chloride ^a	product ^{b,c}
1			 3b: 65% (38%) ^d , 0.5 h
2			 3c: 71%, 0.5 h
3			 3d: 88%, 0.5 h
4			 3e: 74%, 0.5 h
5			 3f: 50%, 4 h
6			 3g: 58%, 0.5 h
7			 3h: 60%, 4 h
8			 3i: 78%, 1 h
9			 3j: 65%, 1 h
10			 3k: 79%, 1 h

^a0.8 equiv of the acid chloride was used. ^bIsolated yield of pure product. ^cUp to 20% of homocoupling was observed. ^dIsolated yield of pure product without catalyst.

(5 mol %) with 4-*tert*-butylbenzoyl chloride (**2b**) within 0.5 h, furnishing the expected diaryl ketone (**3d**) in 88% yield (entry 3). The acylation of the benzylic zinc chloride (**1c**) with the electron-deficient 4-fluorobenzoyl chloride (**2c**) afforded the bis-fluorinated ketone **3e** in 74% yield (entry 4). Remarkably, the readily available benzylic zinc reagent **1d**^{15a} bearing an ester moiety underwent an efficient acylation with 4-chlorobenzoyl chloride (**2a**) leading to the functionalized diaryl ketone **3f** in 50% yield (entry 5).

Additionally, the electron-rich (4-methoxybenzyl)zinc chloride (**1e**) reacted with 4-*tert*-butylbenzoyl chloride (**2b**) to afford ketone **3g** in 58% yield (entry 6). Interestingly, the acylation of the sulfur-containing benzylic zinc chloride (**1f**) with 2-thiophenecarbonyl chloride (**2d**) succeeded within 4 h at 25 °C to give the desired aryl heteroaryl ketone (**3h**) in 60% yield (entry 7). Finally, the reaction of a secondary benzylic zinc reagent (**1g**) with selected acid chlorides (**2a,d,e**) provides the diaryl ketones **3i–k** in 65–79% yield (entries 8–10). In all of these acylations, less than 20% homocoupling of the benzylic moiety was observed.

Additionally, we have extended this cross-coupling to arylzinc chlorides. Compared to benzylic zinc reagents, the carbon–zinc bond of arylzinc halides has a much lower ionic character and, therefore, is significantly less reactive. Thus, the direct acylation of the electron-poor arylzinc halide **4a**, conveniently prepared via an I/Mg exchange using *i*PrMgCl·LiCl²⁶ followed by transmetalation with ZnCl₂, proceeds only sluggishly with 4-chlorobenzoyl chloride (**2a**) at 50 °C in THF providing the ketone **5a** in 34% yield even after a prolonged reaction time of 12 h. In contrast, when FeCl₂ (5 mol %, 99.5% pure) was used, the acylation with **2a** was complete after 4 h reaction time at 50 °C and the desired ketone **5a** was isolated in 62% yield (Table 3, entry 1).²⁷ Similarly, the arylzinc reagent **4a** underwent several acylations with the functionalized acid chlorides **2b,c,f** leading to the diaryl ketones **5b–d** in 65–83% (entries 2–4). Furthermore, the related arylzinc reagent (**4b**), bearing an ester moiety in meta position, reacted rapidly with the acid chlorides **2a,b,h** to afford the ketones **5e–g** within 4 h in 60–75% yield (entries 5–7). Performing such acylations using the ortho-substituted arylzinc chloride (**4c**) in combination with the two electron-poor acid chlorides 4-chlorobenzoyl chloride (**2a**) and 4-cyanobenzoyl

Table 3. Iron-Catalyzed Acylations of Arylzinc Reagents (4a–f) with Acid Chlorides (2a–h)

FG_1 = 2-CO₂Et; 3-CO₂Et; 4-CO₂Et; 4-CF₃; 3-Me; 4-F; 4-OMe
 FG_2 = 4-F; 4-Cl; 4-Br; 4-CN; 4-^tBu; 4-OMe

entry	arylzinc reagent	acid chloride ^a	product ^{b,c}	entry	arylzinc reagent	acid chloride ^a	product ^{b,c}
1			 5a: 62% (56% ^d , 34% ^e), 4 h	9			 5i: 62%, 4 h
2			 5b: 65%, 4 h	10			 5j: 83%, 3 h
3			 5c: 83% (21% ^e), 4 h	11			 5k: 62%, 3 h
4			 5d: 68%, 4 h	12			 5l: 76%, 2 h
5			 5e: 75%, 4 h	13			 5m: 70%, 2 h
6			 5f: 72%, 4 h	14			 5n: 79%, 2 h
7			 5g: 60%, 4 h	15			 5o: 74%, 3 h
8			 5h: 74%, 4 h	16			 5p: 82%, 3 h
				17			 5q: 82%, 3 h

^a0.8 equiv of the acid chloride was used. ^bIsolated yield of pure product. ^cUp to 10% of homocoupling was observed. ^dFeCl₂ with a purity of 99.99% was used. ^eIsolated yield of pure product without catalyst.

chloride (2g) proceeded smoothly at 50 °C, yielding the functionalized diaryl ketones 5h,i in 62–74% yield (entries 8 and 9).

The scope of this acylation procedure proved to be quite general for a range of other functionalized arylzinc reagents. Thus, the reaction of (4-(trifluoromethyl)phenyl)zinc chloride (4d) with acid chlorides bearing electron-donating substituents,

like 4-*tert*-butylbenzoyl chloride (2b) and 4-methoxybenzoyl chloride (2e), gave the desired ketones 5j,k in 62–83% yield (entries 10 and 11). In addition, (4-fluoro-3-methylphenyl)zinc chloride (4e) underwent efficient acylations with the acid chlorides 2a,b,e to afford the functionalized diaryl ketones (5l–n) in 70–79% yield (entries 12–14). Finally, the electron-rich (4-methoxyphenyl)zinc chloride (4f) reacted with selected acid

chlorides bearing a chlorine (**2a**), a *tert*-butyl (**2b**), or a fluorine substituent (**2c**) in the para position leading to the ketones **5o–q** within 3 h at 50 °C in 74–82% yield (entries 15–17).

In summary, we have shown that various benzylic zinc and arylzinc halides react with a broad range of acid chlorides at 25–50 °C leading to polyfunctionalized diaryl and aryl heteroaryl ketones within 0.5–4 h. Further extensions of this methodology are currently underway in our laboratories.²⁸

■ ASSOCIATED CONTENT

● Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: [10.1021/acs.orglett.6b01677](https://doi.org/10.1021/acs.orglett.6b01677).

Detailed experimental procedures and analytical data (PDF)

■ AUTHOR INFORMATION

Corresponding Author

*E-mail: paul.knochel@cup.uni-muenchen.de.

Notes

The authors declare no competing financial interest.

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- (22) For an additive screening, see the [Supporting Information](#).
- (23) For a solvent screening, see the [Supporting Information](#).
- (24) For the use of FeCl₂ of even higher purity (99.99%), see [Table 3](#), entry 1.
- (25) The uncatalyzed reaction proceeds always in ca. 20–30% lower yield as in the presence of 5 mol% of FeCl₂ (see also [Table 2](#), entry 1, or [Table 3](#), entries 1 and 3).
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- (27) Although a yield of 62% of pure product was isolated, indicating that some side reactions may occur, the GC-analysis of the crude reaction mixture showed a clean reaction with only small amounts of side products.
- (28) The Fe-catalyzed acylation of alkylzinc halides proceeds under similar conditions (THF, 25 °C). For preliminary experiments, see the [Supporting Information](#).